

Ratan Lal Bunkar

R&D Hardware Systems Engineer | Connected Health Devices
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Summary

Hardware Systems Engineer (MS from NTUT, BS from IIT) with 4+ years of experience owning system integration for **ISO 11608 and IEC 60601-compliant** connected health platforms. Expert in defining system architecture and managing technical trade-offs across HW/FW/ME from concept through high-volume NPI. Specialized in deep-dive Root Cause Analysis (RCA) for power management and safety-critical performance validation.

Experience

R&D Hardware Systems Engineer

SHL Medical

May 2023 – Present

Taoyuan, Taiwan

- Led system-level integration and verification for 2+ connected electromechanical platforms, supporting compliance with **ISO 11608 and IEC 60601-1**; accelerated design transfer by 15 weeks through structured cross-functional issue resolution.
- Owned end-to-end validation workflows for the Elexy Power Unit, ensuring **100% requirements-to-test traceability** for safety-critical electrical performance, timing behavior, and automated injection logic.
- Designed a sensor-instrumented Mechanical Validation Bench to characterize actuator timing pipelines, reducing injection time variance by **75%** ($\pm 2s \rightarrow \pm 0.5s$) using Design of Experiments on load capacity and viscosity effects.
- Built an SBC-based ATE platform for system V&V, cutting test cycle time by **90%** (**4 min** $\rightarrow$ **<30s**) while ensuring accurate BLE timestamp capture and repeatable evidence generation.
- Resolved critical battery SoC reporting failures via custom UART diagnostics, identifying voltage relaxation and pogo-pin resistance as root causes of erratic 60% SoC jumps.
- Performed cross-domain debugging to mitigate motor overheating risks, introducing firmware sanity checks to reject impossible sensor states and stabilize system behavior.
- Supported defect triage across HW/FW/ME, reproducing corner cases, designing targeted experiments, and delivering structured logs and reports for senior engineering decisions.
- **Programs Supported:** SmartHub (BLE data collector), Elexy (electromechanical autoinjector), Molly cCap (connected module) from prototyping through production ramp.

Research Assistant, Systems & Control Lab

National Taipei University of Technology (NTUT)

Feb 2021 – Jan 2023

Taipei, Taiwan

- Collaborated with a cross-functional research team to develop a novel noise reduction method utilizing ****Non-negative Matrix Factorization (NMF)**** to isolate clean singing voices from complex backgrounds.
- Applied advanced signal processing techniques, including ****RPCA and REpet****, to improve voice extraction accuracy and characterize acoustic data integrity.
- **Publication:** Ratan L. Bunkar, K.Y. Lian, "Nonnegative Matrix Factorization based noise reduction method over singing voices extracted from RPCA and REpet," *International Conference on System Science and Engineering (ICSSE)*, 2022, I.D. 1147.

Control Systems Engineering Intern

DCM Engineering Products

Mar 2019 – Aug 2019

Punjab, India

- Implemented PLC control logic (Siemens Step 7, Ladder Logic) and HMI dashboards; supported commissioning, sensor wiring, signal mapping, and troubleshooting documentation.

Core Skills

Hardware Systems Engineering

Embedded electromechanical integration, System architecture & MBSE (SysML/UML), System Requirements and Design ownership, HW/FW bring-up & interface debugging, Design trade-offs, DFM/DFT, NPD $\rightarrow$ NPI execution

Connectivity & Interfaces

BLE, LTE CAT M1, UART, SPI, Serial (RS-232/422/485), Protocol Analysis, Performance Profiling

System Debug & Reliability

Root cause analysis (HW/FW/ME), Timing/latency margin testing, Log & packet analysis, Oscilloscope-based debugging

Validation Platforms

Custom hardware fixtures & jigs, Embedded Test Fixtures, Python automation, State machines, Data logging & visualization

Embedded & Electromechanical Platforms

Mechatronic Subsystems, Sensor Fusion, Real-Time Control Systems, Hardware Reliability Engineering, Corner Case Analysis, Design of Experiments (DoE) Actuator timing pipelines, Mixed-Signal Debugging, Automated Test Frameworks

Regulatory & Standards

ISO 11608 (Needle-based injection systems), IEC 60601 (Basic safety & essential performance), Design Control (21 CFR 820.30), Risk Management (ISO 14971), HVM-ready Documentation.

Tools & Languages

Python, C/C++, Linux, Git, Wireshark / Pyshark, SysML / UML (Cameo, Enterprise Architect), JIRA / Confluence / Siemens Teamcenter (PLM) / Polarion (ALM/Traceability)

Education

National Taipei University of Technology (NTUT)

Taipei, Taiwan

Master of Science in Electrical Engineering

Feb 2021 – Jan 2023

- **Thesis:** "Noise Reduction Using Non-negative Matrix Factorization with Distributive Penalties on Singing Voices Extracted from RPCA and REpet."
- **Awards:** 2021–2023 Taipei Tech International Student Scholarship, NTUT.
- **Grant:** 2021–2023 SVS Project Research Grant, Systems and Control Lab, NTUT.

Indian Institute of Technology, Ropar

Punjab, India

bachelors of Science in Electrical Engineering

April 2016 – August 2020

- **Thesis:** nRF52-Based BLE Mesh Networking for Tracking Moving Machines, Robots, and Personnel on Shop Floor"
- **Achievements:** 2020 Achievements: Scored Top 20 Percentile in JEE Mains and advanced.